Bit Operation Cost of “Holdout” Key-Recovery Attacks Against Classic McEliece

Markku-Juhani O. Saarinen

Tampere University, Finland
markku-juhani.saarinen@tuni.fi

Revision: 30 August 2026

Abstract. We cost the best Holdout key-recovery attack that we currently know for each Classic McEliece parameter set. One shortened coordinate set of the direct locator-recovery method of Ghoshal, Ishai, Jain, and Sun suffices for every set: public parity-check elimination selects the $mt + 1$ compatible labels needed for key completion and replaces the four or five shortened sets used in the source analysis. We make every sparse linear-system solve reliable by applying Eberly’s random diagonal scaling and iterative scalar Lanczos solver over a larger field, mapping each returned solution back to the original field, and substituting it into the original equations. Separately costing matrix passes that use only binary derivative data and operations with general field coefficients gives complete conditional upper ledgers of $2^{126.77}$, $2^{145.22}$, $2^{137.48}$, $2^{136.65}$, and $2^{137.48}$ bit operations in parameter-set order. Their simultaneously stored state is at most $2^{51.33}$, $2^{59.10}$, $2^{55.18}$, $2^{54.86}$, and $2^{55.18}$ bits. All five work estimates lie below the NIST classical-gate reference level for their claimed category. The estimates remain conditional on the source’s conjectures about the canonical form of binary Goppa codes and the rigidity of rank-one solutions, as well as its predicted matrix ranks. They are arithmetic and state estimates, not elapsed-time claims, and no Classic McEliece target key has been recovered. *Note: This is a living document: the estimate, assumptions, and scope will be updated as algorithms and reproducible evidence improve.*

Keywords: Classic McEliece · Holdout Attack · Key Recovery · Bit Operations · Sparse Linear Algebra

1 Introduction

Classic McEliece derives its public code from a binary Goppa code [3,9]. Recent Holdout and higher-order-vanishing constructions turn publicly computable polynomial relations into candidate support values, called *locators*, and then conditionally into an equivalent decoding key [5,6,7,11]. Their concrete cost is sensitive to three details that an algebraic-operation count can hide: how many shortened coordinate sets must be processed, which matrix entries are binary, and whether the algorithm used to solve the sparse linear systems has a quantified failure probability.

Some Terminology. Here a *holdout* is a public coordinate deliberately omitted from a set of vanishing constraints so that the resulting relations can reveal local information at that coordinate. A *projective label* is a proposed locator in the projective line $\mathbb{P}^1(\mathbb{F}_q)$, consisting of the q field elements and one point at infinity. The direct route repeatedly tests such labels at held-out coordinates; accepted labels are then passed to a Goppa-key completion algorithm and full public-key verification.

Current Revision. This paper resolves those three costing questions for the direct locator-recovery route in the 27 August 2026 revision of Ghoshal et al. [5]. We report one best current-route total for each standardized parameter set. The earlier singleton-tangent route (studied in previous revisions of this report) is not retained as a competing model because it costs more while leaving additional attack stages unresolved. That earlier route still requires methods to derive tangent data from the public key, reject incorrect locator guesses, align coordinates that differ by the Frobenius field automorphism $x \mapsto x^2$, and extend a partially recovered support. Necessary conditions in Vedenev's method for retaining more coordinates after shortening also exclude the small dimensions on which the earlier favorable cost depended. Section 6 records these short rejection arguments without preserving superseded numerical tables.

Our contributions:

1. a proof that one shortened coordinate set supplies the $mt + 1$ mutually compatible labels needed by the key-completion algorithm for every Classic McEliece parameter set;
2. a reliable method for solving the affine sparse systems over a larger field, including an exact linear map back to the original field and substitution into the original equations;
3. an explicit decomposition into binary operations and general field operations, with a finite cost for every remaining algorithmic step; and
4. a small self-contained artifact, `mccost.py`, that recomputes only the dimensions, reliability choices, state bounds, and five current-route totals in this paper.

The public artifact is available at `github.com/mjosaarinen/mccost`.

The calculation is conditional in the precise sense of Section 7; in particular, the paper does not assign probabilities to unproved structural conjectures.

2 The Direct Locator-Recovery System

Let m be the extension degree, $q = 2^m$ the field size, n the code length, t the degree of the secret Goppa polynomial, and k the public-code dimension. The National Institute of Standards and Technology (NIST) post-quantum evaluation criteria use 2^{143}, 2^{207}, and 2^{272} classical gates as reference levels for Categories 1, 3, and 5, respectively [10]. These are reference comparisons for the bit-operation metric used here; they are not simply 2^{128}, 2^{192}, and 2^{256} elementary operations.

Table 1 gives the five fixed parameter cells from the source analysis. *Shortening* on ℓ coordinates means restricting to codewords that are zero there and then deleting those coordinates; it gives the remaining length and dimension $(n_\ell, k_\ell) = (n-\ell, k-\ell)$. A polynomial is *homogeneous* when all of its monomials have the same total degree. A Hermite jet at a point is the collection of a polynomial's value and its Hasse derivatives, the coefficient-based derivatives appropriate in finite characteristic, up to a chosen order; multiplicity s means imposing vanishing for derivative orders below s. In the profile column, s^a means that a points receive multiplicity s. We use the source's cells without re-optimizing them.

Table 1. Direct locator-recovery cells and NIST classical-gate reference levels.

Parameter set	Category / reference	(m, n, t, k)	(ℓ, d)	(n_ℓ, k_ℓ)	Profile
mceliece348864	1 / 2^{143}	$(12, 3488, 64, 2720)$	$(2569, 7)$	$(919, 151)$	$5^{42}6^{877}$
mceliece460896	3 / 2^{207}	$(13, 4608, 96, 3360)$	$(3159, 8)$	$(1449, 201)$	$5^{4}6^{177}7^{1268}$
mceliece6688128	5 / 2^{272}	$(13, 6688, 128, 5024)$	$(4802, 7)$	$(1886, 222)$	$5^{35}6^{1851}$
mceliece6960119	5 / 2^{272}	$(13, 6960, 119, 5413)$	$(5198, 7)$	$(1762, 215)$	$5^{24}6^{1738}$
mceliece8192128	5 / 2^{272}	$(13, 8192, 128, 6528)$	$(6306, 7)$	$(1886, 222)$	$5^{35}6^{1851}$

For a detailed derivation, we use mceliece348864 as the running example. Its parameters are

$$(m, n, t, k) = (12, 3488, 64, 2720). \tag{1}$$

Its fixed cell has $(\ell, d) = (2569, 7)$, $(n_\ell, k_\ell) = (919, 151)$, and profile $5^{42}6^{877}$. The same formulas below are evaluated at every row of Table 1 by the artifact.

At a high level, the construction forms two systems of homogeneous polynomial values and derivatives from the shortened binary public generator. For each proposed locator value, rows that fix otherwise arbitrary scaling and rows that test that locator turn the two systems into an affine feasibility test $Ax = b$. The structural premises stated in Section 7 connect accepted solutions to the hidden Goppa locators; the present section only counts the resulting matrices.

For multiplicity s, one homogeneous jet copy has $\binom{k_\ell+s-1}{s-1}$ columns. Thus the profile gives

$$M = 42\binom{155}{4} + 877\binom{156}{5} = 633{,}864{,}594{,}132. \tag{2}$$

The space of homogeneous degree-seven equations has

$$N_d = \binom{k_\ell + d - 1}{d} = \binom{157}{7} = 407{,}340{,}975{,}756. \tag{3}$$

The four blocks that fix scaling each have length $L_5 = \binom{155}{4} = 23{,}130{,}030$. The largest block that tests a proposed locator is

$$B = \binom{152}{2} + \binom{153}{3} + \binom{154}{4} + \binom{155}{5} = 721{,}656{,}784. \tag{4}$$

Consequently, the affine feasibility matrix has

$$U = 2M = 1{,}267{,}729{,}188{,}264, \tag{5}$$
$$R = 2N_d + 4(L_5 - 1) + 1 + B = 815{,}496{,}128{,}413. \tag{6}$$

The two value-and-derivative systems are binary: the shortened public generator and all their zero-one incidence coefficients, which record the relevant monomial–derivative containments, lie in the two-element field $\mathbb{F}_2$. A direct count from the multiplicity profile gives

$$z_{\text{bin}} = 2N_d \left(42 \sum_{r=0}^{4} \binom{7}{r} + 877 \sum_{r=0}^{5} \binom{7}{r} \right) = 89{,}124{,}576{,}131{,}509{,}776 \tag{7}$$

nonzero binary entries. Only the scaling-fixing and locator-testing pieces require arbitrary coefficients from the base field $\mathbb{F}_{2^{12}}$; their two copies have

$$z_{\text{gen}} = 2\big(4(L_5 - 1) + 1 + B\big) = 1{,}628{,}353{,}802 \tag{8}$$

nonzero entries. Treating both populations as general field multiplications would therefore discard the main circuit-level distinction.

The scan performs one preliminary test and then tests n_ℓ locator positions against all $q+1$ elements of the projective line. We conservatively charge every test as one system of the largest size:

$$S = (n_\ell + 1)(q + 1) = 3{,}769{,}240. \tag{9}$$

3 Why One Shortened Chart Suffices

The direct construction labels every position of one shortened coordinate set. Its shortening gate verifies that the deleted public columns have full rank $|I|$, so the shortened code has dimension $k - |I|$. The source analysis covers the entire public support with four or five such sets, but the known-point key-completion algorithm needs only $mt + 1$ labels expressed in one common projective coordinate system [8]. The following public reduction supplies them in every row of Table 1.

Lemma 1 (One-chart selection). *Let I be the shortening set, let $J = [n] \setminus I$ be the remaining positions, and let H be a full binary parity-check matrix, so the public code is* $\ker H$*. If the shortened systematic construction has dimension $k - |I|$ and $|J| \geq n - k + 1$, then* $\operatorname{rank} H_J = n - k = mt$*. Hence the recovered labels on J contain mt positions whose parity-check columns are linearly independent and one further labelled position.*

Proof. Extending a shortened word by inserting zeros at I identifies $\ker H_J$ with the public code shortened on I. This kernel has dimension $k - |I|$. The rank–nullity theorem gives

$$\operatorname{rank} H_J = |J| - (k - |I|) = n - k = mt.$$

Gaussian elimination over $\mathbb{F}_2$ publicly selects mt independent columns. The inequality $|J| \geq mt + 1$ leaves at least one further labelled position.

Table 2. Capacity of one shortened chart for known-point completion.

Parameter set	n_ℓ	mt	Required $mt+1$	Surplus
mceliece348864	919	768	769	150
mceliece460896	1,449	1,248	1,249	200
mceliece6688128	1,886	1,664	1,665	221
mceliece6960119	1,762	1,547	1,548	214
mceliece8192128	1,886	1,664	1,665	221

To apply an affine recovery algorithm, one projective value must be designated as the point at infinity, or *pole*. For each projective value absent from the selected labels, the completion step chooses it as the pole, converts the other $mt + 1$ labels to affine field values, applies the degree-t Goppa-polynomial recovery algorithm, extends the candidate support, and accepts only a complete public-key verification. One trial necessarily chooses the correct pole. We charge every possible pole with a deliberately loose finite bound in Section 5; this completion work is smaller than the locator scan in every row. Thus one chart uses $(n_\ell{+}1)(q{+}1)$ largest-system equivalents, with no multiplier for processing several shortened coordinate sets.

4 A Reliable Affine Sparse Solve

Write $\mathbb{F}_{2^m}$ for the field used by the public construction. Each candidate asks whether the affine linear system

$$Ax = b, \qquad A \in \mathbb{F}_{2^m}^{R\times U}, \quad b \in \mathbb{F}_{2^m}^{R}, \tag{10}$$

is consistent, meaning that some x satisfies every equation. The matrix is sparse because almost all of its entries are zero. A heuristic solver is inadequate for a headline ledger: millions of trials amplify even a small probability of failing on a solvable system. We instead use Eberly's reliable scalar Lanczos construction [4]. Scalar Lanczos is an iterative linear-system algorithm that accesses a matrix through products with vectors. Eberly first multiplies rows and columns by random nonzero diagonal entries; this *preconditioning* makes the required rank and solver properties fail only with a bounded probability.

Choose δ divisible by m and write $\mathbb{F}_{2^\delta}$ for the resulting field $\mathbb{F}_{2^\delta}$. Then $\mathbb{F}_{2^\delta}$ contains a unique subfield of size 2^m, which we identify with $\mathbb{F}_{2^m}$. Sample independent diagonal matrices $D_\alpha \in \mathbb{F}_{2^\delta}^{U\times U}$ and $D_\beta \in \mathbb{F}_{2^\delta}^{R\times R}$ with uniformly random nonzero diagonal entries, and sample a uniform vector $z \in \mathbb{F}_{2^\delta}^{U}$. Define

$$G = D_\alpha A^T D_\beta^2 A D_\alpha \tag{11}$$

and apply scalar Lanczos to

$$Gy = D_\alpha A^T D_\beta^2 (b - Az). \tag{12}$$

Return $x = z + D_\alpha y$. A *retraction* is an $\mathbb{F}_{2^m}$-linear map from $\mathbb{F}_{2^\delta}$ back to $\mathbb{F}_{2^m}$ that fixes every element of $\mathbb{F}_{2^m}$; apply it to every coordinate of x. Finally, perform *exact replay*: substitute the retracted vector into the original equation (10) and refuse it unless equality holds.

Theorem 1 (Reliable affine reduction). *If* (10) *is consistent, the right-hand side of* (12) *is the image under G of a uniformly random vector, which is the input distribution required by Eberly's scalar-Lanczos theorem. When the random diagonal scaling preserves the required rank and Lanczos succeeds, the returned vector solves* (10). *An inconsistent or corrupted output cannot be accepted by exact replay. Moreover, a solution over $\mathbb{F}_{2^\delta}$ can be mapped to a solution over $\mathbb{F}_{2^m}$ by a fixed $\mathbb{F}_{2^m}$-linear retraction.*

Proof. Fix one solution x_0 of (10). Its randomized right-hand side is

$$D_\alpha A^T D_\beta^2 (b - Az) = G D_\alpha^{-1}(x_0 - z).$$

Since z is uniform, the hidden preimage is uniform in $\mathbb{F}_{2^\delta}^{U}$. This is exactly the image-of-a-random-vector distribution in Eberly's theorem. When $\operatorname{rank} G = \operatorname{rank} A$, the kernels of G and AD_α agree. Hence any solution y of (12) satisfies $A(z + D_\alpha y) = b$.

Because $m \mid \delta$, $\mathbb{F}_{2^\delta}$ is an extension of $\mathbb{F}_{2^m}$. Choose an $\mathbb{F}_{2^m}$-linear map $\pi : \mathbb{F}_{2^\delta} \to \mathbb{F}_{2^m}$ with $\pi(1) = 1$, and apply it coordinatewise. It fixes $\mathbb{F}_{2^m}$, so $A\pi(x) = \pi(Ax) = \pi(b) = b$. Finally, exact replay rejects every vector that does not satisfy the original system, regardless of how it arose.

For a matrix with at most U columns and rank at most R, Eberly's bounds give, per system,

$$p_{\text{pre}} \le \frac{11U^2 - U}{2(2^\delta - 1)}, \qquad p_{\text{L}} \le \frac{R(R+1)}{2^\delta}. \tag{13}$$

Here p_{pre} bounds failure of the random diagonal scaling and p_{L} bounds failure of scalar Lanczos. Adding these probabilities for every one-chart system gives a valid upper bound even when the failure events are not independent. For each parameter set, the artifact selects the least-work multiple of m satisfying $S(p_{\text{pre}} + p_{\text{L}}) < 2^{-128}$. For the running example this gives

$$\delta = 240, \qquad \log_2\bigl(S(p_{\text{pre}} + p_{\text{L}})\bigr) < -135.1793. \tag{14}$$

This is a bound on algorithmic solver failure. It does not measure the truth of the structural assumptions in Section 7.

5 Bit-Operation Ledger

Represent each element of $\mathbb{F}_{2^\delta}$ by the δ binary coefficients of a polynomial. The direct, or schoolbook, multiplication circuit uses at most δ^2 Boolean AND operations, $\delta^2-(2\delta-1)$ exclusive-or (XOR) operations to combine equal-degree products, and $\delta(\delta-1)$ XOR operations to reduce the result modulo the fixed irreducible polynomial that defines the field. We therefore charge $C_\times$ for one multiplication and C_{ma} for one multiplication followed by an addition:

$$C_\times(\delta)=3\delta^2-3\delta+1, \qquad C_{\mathrm{ma}}(\delta)=3\delta^2-2\delta+1. \tag{15}$$

At $\delta = 240$, these are 172,081 and 172,321 bit operations. This transparent circuit upper bound does not use an assumed multiplier for an unspecified field implementation.

One multiplication of a vector by the preconditioned matrix G in (11) is charged by

$$C_G = 2z_{\mathrm{bin}}\delta + (2z_{\mathrm{gen}}+2U+R)C_{\mathrm{ma}}(\delta) = 43{,}357{,}797{,}573{,}481{,}659{,}425. \tag{16}$$

The first term consists only of XOR operations on bit vectors. The second conservatively charges the arbitrary field coefficients and every diagonal scaling as a full multiply-add. Between matrix products, scalar Lanczos updates a fixed collection of vectors and inner products; we bound one such recurrence step and the combined initialization and finalization by

$$C_{\mathrm{rec}} = (17U+4\delta+3)C_{\mathrm{ma}}(\delta), \tag{17}$$

$$C_{\mathrm{init}} = (20U+4\delta+3)C_{\mathrm{ma}}(\delta). \tag{18}$$

Substituting one candidate into the original base-field equations costs at most

$$C_{\mathrm{replay}} = z_{\mathrm{bin}}m + z_{\mathrm{gen}}C_{\mathrm{ma}}(m) + Rm. \tag{19}$$

Generating the random diagonal entries and affine shift, then applying a general m-by-δ binary linear map to every output coordinate, costs at most

$$C_{\mathrm{source}} = (2U+R+mU)\delta. \tag{20}$$

These costs are included in each feasibility-system row of Table 3. Charging $R+3$ multiplications by G and R recurrence steps gives

$$C_{\mathrm{sys}} = (R+3)C_G + RC_{\mathrm{rec}} + C_{\mathrm{init}} + C_{\mathrm{replay}} + C_{\mathrm{source}} < 2^{104.92}. \tag{21}$$

The Gaussian elimination used for the public one-chart selection is bounded by

$$C_{\mathrm{select}} = n_\ell(n-k)^2 + n_\ell(n-k) < 2^{29.02}. \tag{22}$$

Put $h = mt = 768$ for the number of parity-check rows and $L = 769$ for the number of selected labels. Enumerating every possible missing pole and, conservatively, permitting even a field inversion for every algebraic operation gives the finite ceiling

$$C_{\mathrm{finish}} \le (q+1-L)(qL^6+qnh^4)\,2mC_\times(m) < 2^{94.45}. \tag{23}$$

The key-completion work is well below the locator scan and changes no reported headline digit. The complete conditional ledger is therefore

$$C_{\text{total}} = SC_{\text{sys}} + C_{\text{select}} + C_{\text{finish}} < 2^{126.77} \text{ bit operations.} \tag{24}$$

Table 3. Detailed conditional attack ledger for mceliece348864.

Item	Bit-operation upper bound
One affine feasibility system	$2^{104.92}$
All 3,769,240 systems	$2^{126.77}$
Public label selection	$2^{29.02}$
Projective completion and verification	$2^{94.45}$
Complete conditional ledger	$\mathbf{2^{126.77}}$

Applying the same formulas to every fixed cell gives Table 4. Work is measured in bit operations and state in simultaneously stored bits. The solver-failure column is the union bound over every charged affine system; the margin subtracts the work exponent from the NIST classical-gate reference exponent for the claimed category.

Table 4. Best known conditional Holdout attack ledger for every Classic McEliece parameter set.

Parameter set	Category / reference	Solver field	Work	State	Solver failure	Margin
mceliece348864	1 / 2^{143}	$\mathbb{F}_{2^{240}}$	$2^{126.77}$	$2^{51.33}$	$< 2^{-135.18}$	16.23
mceliece460896	3 / 2^{207}	$\mathbb{F}_{2^{260}}$	$2^{145.22}$	$2^{59.10}$	$< 2^{-138.20}$	61.78
mceliece6688128	5 / 2^{272}	$\mathbb{F}_{2^{247}}$	$2^{137.48}$	$2^{55.18}$	$< 2^{-132.52}$	134.52
mceliece6960119	5 / 2^{272}	$\mathbb{F}_{2^{247}}$	$2^{136.65}$	$2^{54.86}$	$< 2^{-133.25}$	135.35
mceliece8192128	5 / 2^{272}	$\mathbb{F}_{2^{247}}$	$2^{137.48}$	$2^{55.18}$	$< 2^{-132.52}$	134.52

The *active state* is the information that must be stored simultaneously. For each row the bound retains eight vectors of length U for the recurrence and random shift, two work vectors of length R, and the z_{gen} arbitrary field coefficients. For the running example,

$$C_{\text{state}} = \big(8U + 2R + z_{\text{gen}}\big)\delta < 2^{51.33} \text{ bits.} \tag{25}$$

This is a live-state upper bound for the mathematical schedule, not a claim that the computation fits a particular machine. Memory traffic, communication, checkpointing, and redundant execution are not converted into additional bit operations.

6 Why Other Proposed Savings Do Not Change the Tally

Earlier one-coordinate route. Previous revisions priced a route that built a separate relation space after omitting each one of several selected coordinates. It then attempted to recover derivative information at those coordinates, guess some locator values, align the resulting values across Frobenius automorphisms, and interpolate the hidden curve. Its lowest numerical subtotal was larger than Table 3 and omitted more unresolved work: constructing the required public relation spaces, rejecting an incorrect tuple of locator guesses, extending the correct partial support, and aligning coordinates were still conditional. The 30 August 2026 revision of Vedenev's method for retaining additional curve coordinates after shortening also requires $k_\ell \geq 620$ at its first parameter choice satisfying the necessary inequalities, excluding the small shortened dimensions on which that route's favorable cost depended [11]. It therefore cannot improve the present headline, and we do not reproduce the obsolete scan or tables. Apon's algebraic-geometric obstruction further explains why a few guessed locators alone cannot identify the binary-Goppa curve [1].

Reconstructing only part of a Wiedemann solution. The Wiedemann method solves sparse linear systems by generating sequences of matrix–vector products. Chou and Niederhagen optimize its use in extended linearization (XL) by reconstructing only the selected coordinates that encode their candidate solution [2]. The direct Holdout route instead needs every coordinate of a relation vector to check its derivatives and continue key recovery, so requesting all coordinates restores a full reconstruction pass. Choosing different numbers of vectors on the left and right can reduce one constant in the sparse matrix passes, but no complete-output reliability and state analysis is currently available for this matrix shape. We therefore make no change to Table 3.

7 What the Estimate Does and Does Not Establish

The following parts of all five ledgers in Table 4 are proved or given explicit finite upper bounds in this paper:

1. the exact fixed dimensions and the separation of nonzero entries into binary and arbitrary-field entries;
2. the rank-mt one-chart selection lemma and the $mt + 1$ label-capacity check for every parameter set;
3. the randomized conversion of each affine system to the form required by Eberly's reliable solver theorem;
4. the linear map back to the base field, exact substitution into the original system, and the aggregate solver-failure bound;
5. polynomial-basis circuit costs, the $mt + 1$-label key-completion ceilings, and the active-state bounds.

The route remains conditional on:

Assumption 1 (Imported structural premises) *The canonical form conjecture for binary Goppa codes and the rank-one rigidity conjecture of Ghoshal et al. hold for the target. Their predicted ranks for the complete matrices and the matrices restricted to selected coordinates also hold for the applicable parameter choice in Table 1.*

No probability is assigned to Assumption 1. The solver-failure bounds in Table 4 are conditional on those premises and should not be read as end-to-end attack success probabilities. No target-scale implementation of the full locator scan has produced public relations, locators, a Goppa polynomial, or a Classic McEliece key. Subject to that boundary, the best attacks we currently know how to tally lie below the corresponding NIST classical-gate reference level for all five Classic McEliece parameter sets.

Acknowledgement

The authors wish to acknowledge CSC – IT Center for Science, Finland, for computational resources.

Use of Artificial Intelligence

These results originate from a systematic effort to implement and improve cryptanalytic attacks on Post-Quantum Cryptography with AI help. We used OpenAI GPT-5.6-Sol and Claude Opus 5 for theory formulation, proofs, analysis, and preparing and executing experiments. Furthermore, we used the LLMs and Grammarly to prepare and improve the presentation.

References

1. Apon, D.: An algebraic-geometry lower bound against the ePrint:2026/1747 McEliece key-recovery attack. Cryptology ePrint Archive, Paper 2026/1810 (2026), `https://eprint.iacr.org/2026/1810`
2. Chou, T., Niederhagen, R.: A practical optimization for wiedemann XL. Cryptology ePrint Archive, Paper 2026/1787 (2026), `https://eprint.iacr.org/2026/1787`
3. Classic McEliece Team: Classic mceliece: conservative code-based cryptography: guide for security reviewers (October 2022), `https://classic.mceliece.org/mceliece-security-20221023.pdf`, document of official Classic McEliece web site.
4. Eberly, W.: Reliable krylov-based algorithms for matrix null space and rank. In: Gutierrez, J. (ed.) Symbolic and Algebraic Computation, International Symposium ISSAC 2004, Santander, Spain, July 4-7, 2004, Proceedings. pp. 127–134. ACM (2004). `https://doi.org/10.1145/1005285.1005305`
5. Ghoshal, A., Ishai, Y., Jain, A., Sun, N.: Quasipolynomial cryptanalysis of the McEliece cryptosystem (or: PIR meets McEliece). Cryptology ePrint Archive, Paper 2026/1630 (2026), `https://eprint.iacr.org/2026/1630`

6. Hemmert, T.: Key recovery for the McEliece cryptosystem using higher-order vanishing. Cryptology ePrint Archive, Paper 2026/1339 (2026), https://eprint.iacr.org/2026/1339
7. Hemmert, T., Wiemers, A.: Distinguishing Goppa codes using higher-order vanishing. In: Heninger, N., Rosulek, M. (eds.) Advances in Cryptology – CRYPTO 2026. 46th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 17–20, 2026, Proceedings, Part IV. Lecture Notes in Computer Science, vol. 16803, pp. 191–223. Springer (2026). https://doi.org/10.1007/978-3-032-35398-6_7, https://eprint.iacr.org/2025/1661
8. Kirshanova, E., May, A.: Breaking goppa-based McEliece with hints. Inf. Comput. **293**, 105045 (2023). https://doi.org/10.1016/J.IC.2023.105045, https://eprint.iacr.org/2022/525
9. McEliece, R.J.: A public-key cryptosystem based on algebraic coding theory. Progress Report DSN PR 42-44, Deep Space Network (1978), https://ipnpr.jpl.nasa.gov/progress_report2/42-44/44N.PDF
10. NIST: Submission requirements and evaluation criteria for the post-quantum cryptography standardization process. NIST Call for PQC Standardization Candidates (December 2016), https://csrc.nist.gov/CSRC/media/Projects/Post-Quantum-Cryptography/documents/call-for-proposals-final-dec-2016.pdf
11. Vedenev, K.: Extending distinguishing to key recovery for subfield subcodes of GRS codes. Cryptology ePrint Archive, Paper 2026/1747 (2026), https://eprint.iacr.org/2026/1747

A Artifact Reproduction

Run

```
python3 mccost.py
```

with Python 3. This command recomputes all five current-route reports and checks their exact arithmetic. The artifact intentionally contains no superseded parameter scans or speculative experiments.

B Changelog

2026-08-24. Initial release, followed by same-day cleanup and clarification.

2026-08-27. Updated the Introduction and asymptotic discussion for the then-current revisions by Vedenev and by Ghoshal et al.; added the proof for the rank of one point's derivative-row block, exact padding of rectangular systems, active-state accounting, joint parameter optimization, and normalization by projective linear transformations. The artifact then separately reconstructed the published conditional competing baseline without importing its unitemized optimized claim.

2026-08-29. Updated for Vedenev's revision dated 2026-08-28: adopted its rank test for rejecting incorrect guesses and the corresponding sensitivity based on the probability that a random matrix loses rank. Selected parameter choices and attack subtotals were unchanged.

2026-08-30. Sections 2–5 now give the one-chart direct locator-recovery route, reliable larger-field solves followed by exact maps to the original fields, and one best current estimate for each of the five parameter sets, compared with the cited NIST classical-gate reference levels. Superseded singleton tables and artifact functions were removed; Section 6 reflects Vedenev's current feasibility boundary, and Section 7 records the revised scope and remaining assumptions.